



STANDARD COLLEGE NTUNGAMO

S.5 PHYSICS PAPER 1

END OF TERM 3, 2025

2hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

The paper has four sections, **A**, **B**, **C** and **D**

Answer **FIVE** questions from this paper.

Section **A** and **B** are *compulsory*, choose **only two** items from section **C** and **only one** item from section **D**

Assume where necessary:

Acceleration due to gravity, g = $9.81ms^{-2}$

Permittivity of free space, ϵ_0 = $8.85 \times 10^{-12}Fm^{-1}$

The constant, $\frac{1}{4\pi\epsilon_0}$ = $9.0 \times 10^9F^{-1}m$

Electronic Charge e = $1.6 \times 10^{-19}C$

Density of air = $1.29kgm^{-3}$

SECTION A (compulsory)

ITEM 1

During a field study, a group of S.5 students visited a rural community where water is collected from a narrow tube fixed in the ground. The students observed the following;

☐ When small insects such as water striders landed on the surface of the water in the basin, they were able to walk and rest on it without sinking.

☐ When the water in the basin was warmed by placing it under the sun for a long time, the insects began to sink more easily.

TASK:

As a learner of physics,

- a) Explain why the insects were able to walk on the surface of the cold water without sinking.
- b) Describe what happened to the surface tension when the water was warmed and explain why insects began to sink.
- c) Water rises to a height, h_0 inside a clean glass capillary tube of radius **0.2 mm** when the tube is placed vertically inside a beaker. Calculate the value of h if the surface tension of water is $7.0 \times 10^{-2} \text{ Nm}^{-1}$ and the angle of contact is 0° .

SECTION B (compulsory)

ITEM 2

A science laboratory stores three different radioactive sources labelled A, B and C. The lab technician wants to use them for different experiments. The following observations were made:

- ☐ When placed near paper, source A is stopped completely.
- ☐ Source B passes through paper but is stopped by a thin sheet of aluminium.
- ☐ Source C passes through aluminium but is greatly reduced by a thick block of lead.

TASK:

As a learner of physics,

- a) Identify the type of radiations emitted by source A, B and C.
- b) State two properties of each type of radiation identified above.
- c) Give four dangers associated with exposure to ionization radiation.
- d) State four nuclear radiation detectors and describe the operation of the ionization chamber.

SECTION C

Do only two items from this section

ITEM 3

- (a) Imagine you are on a small sailboat in the Atlantic Ocean, several miles from any landmass. The Sun is shining, but there are no land marks to help you determine your position and the Sun's altitude.

TASK:

- (i) What instrument would be used to determine the sun's altitude?
 - (ii) Derive the principal of working of the instrument named
 - (iii) Explain how the instrument named performs its function.
- (b) A fish is **3.0m** below the surface of a pond and **2.5m** from the bank. Aman **2.0m** tall stands **4.0m** from the edge of the pond. As the man approached the pond, the fish went into hiding as soon as the man saw it. The man wondered how the fish could see him that quickly, however his colleague told him that the fish is able to see very thing in water an above water at the same time.

HINT: refractive index of water is **1.33**.

TASK: As a physics student, help the man to know:

- (i) How the fish is able to see every thing in water and its surroundings.
- (ii) The distance the man moved towards the edge of the pond for him to see the fish.

ITEM 4

To study behaviour of waves, students placed a small piece of cork on the surface of a swimming pool with calm water, the pool is located at about **100m** from a large building. A stick with a small circular end is dipped in water continuously at regular intervals to generate ripples on water surface. The cork is seen to oscillate up and down at the same point as the ripples pass it. The cork is **2m** from a plastic swimming floater that had been left in the pool, it is observed that the ripples reach the floater 10seconds after passing the cork. The ripples have a wavelength of 15mm and amplitude of 5mm.

During the study, the teacher clapped his hands to call for attention of a group that was discussing. The students were surprised to hear the teachers clap twice.

TASK: As a physics student,

- (i) Explain why the cork vibrates up and down from the same point.
- (ii) Determine the speed at which the ripples move on the pool surface.
- (iii) Determine the frequency of vibration of the cork, and explain how this relates to frequency of the ripples.
- (iv) Find the displacement equation of the ripples, hence find the maximum speed of the cork during its vibration.
- (v) Explain why two claps are heard by the students.
- (vi) Help the students to utilise the scenario of two claps to determine the speed of sound in air.

ITEM 5

In Nyakafunzo village in western Uganda, medical workers received a new solar-powered refrigerator to store vaccines at the local health centre. The effectiveness of vaccines depends on precise temperature control within a narrow range typically (**2 °C to 8 °C**). The built-in digital thermometer frequently fails due to poor quality sensors. A volunteer physics student from a nearly A 'level school suggested the use of platinum resistance thermometer. They set up the circuit to measure the temperature inside the fridge. The student measures the resistance of the platinum wire in the thermometer to calculate the temperature of the fridge using thermodynamic temperature scale and at some point, **$R_0 = 107.7\Omega$** inside the refrigerator

Hint: The resistance **R** of the platinum wire varies with temperature **θ** according to the linear relation **$R_\theta = R_0 (1 + \alpha\theta)$** where **$R_\theta$** is the resistance at temperature, **θ** , **$R_0 = 100\Omega$** is the resistance at 0 °C and **$\alpha = 3.85 \times 10^{-3}$** is the temperature coefficient of resistance of platinum wire.

TASK:

As a learner of physics;

- a) Calculate the temperature inside the refrigerator on thermodynamic scale
- b) Describe steps taken to set up a thermodynamic scale on the above thermometer.
- c) Explain why the platinum resistance thermometer is suitable for monitoring temperatures in sensitive storage like vaccines.

- d) Identify and explain two limitations of using platinum resistance thermometers in rural Uganda settings
- e) Give three other practical applications of platinum resistance thermometers in different fields in Uganda.

SECTION D

Do one item from this section

ITEM 6

A S5 class presented electrostatics experiments expo to the entire school. During the expo, Alice combed her hair and used the comb to pick up small paper bites. Alice explained to the audience that the comb acquires charge during combing. During the same expo, Patience stood on a wooden stool and held on the dome (hollow metal sphere) of a Van de Graaff generator with both her hands, the dome is of **2m** diameter and holds a charge of **$+5.0\mu C$** . Her hair stood like spikes on her head. These observations amused the audience and had many unanswered questions.

TASK:

As a physics student,

- (i) Explain the observations during the expo.
- (ii) Explain how a Van de Graff generator works.
- (iii) Find the electric potential and electric field intensity at the surface of the sphere.

ITEM 7

A keyboard designer is creating a new keyboard where each key acts as a capacitor. The initial unpressed separation between the capacitor plates is **5.2mm** and the plate area is **$1.6 \times 10^{-4}m^2$** . The capacitor is filled with air; the designer requires that pressing a key causes the capacitance to increase by **0.2pF** to trigger the signal due to decrease in separation. The initial capacitance of the capacitor has to be determined before the capacitor is fixed to make the key but the designer does not know how to test capacitance.

TASK:

As a physics student,

(a) help the designer to know;

- (i) The initial capacitance of the key in its unpressed state.
 - (ii) The final distance between the plates when the key is pressed.
 - (iii) How capacitance can be determined experimentally before the capacitor is used in the keyboard if a vibrating reed switch is provided.
- (b) Explain the effect on capacitance if the space between the plates is filled with a dielectric.

END

***The physics dep't wishes you a merry X-mas and
prosperous New Year 2025***